

HumanEd: Mo-6 "Morris" 6-Axis Serial Manipulator

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1 Overview

The Mo-6 robot has been designed as a low-cost 6-axis serial manipulator that can be easily tailored to use different control systems and end-effectors. The robot follows the standard joint arrangement seen on many industrial 6-DOF manipulators, with a 2R open-chain attached to a rotating base used as the positioning system for a spherical wrist.

1.1 Kinematics

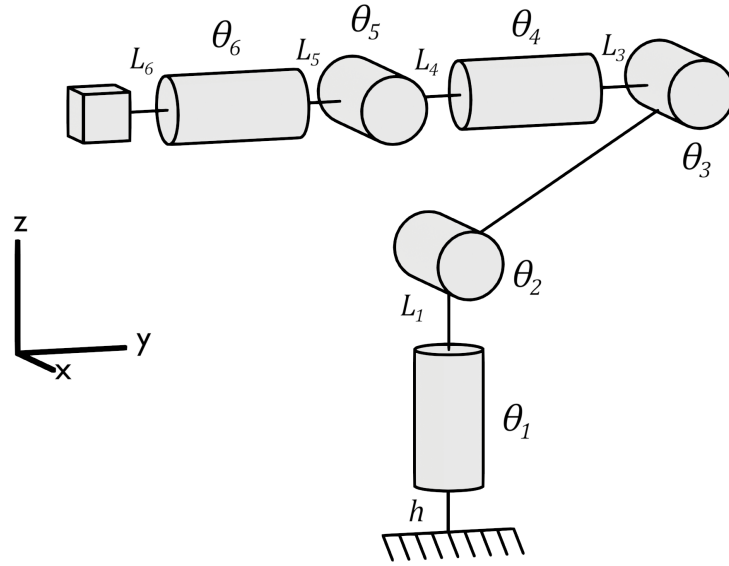


Figure 1: Joint diagram of the Mo-6 robot.

Figure 1 shows the joints of the Mo-6 robot. θ_1 , θ_2 , and θ_3 comprise the positioning joints; θ_4 , θ_5 , and θ_6 comprise the orienting joints, constructed in the spherical wrist configuration.

The only notable deviation from this structure on the Mo-6 is that when the robot is in its rest position, there is an angled offset between joint 2 and joint 3. This can be seen in Figure 2, which shows a side view of Mo-6 with the aesthetic shell of its second link removed. The offset has been marked with distances L_{2y} and L_{2z} , and θ as the resulting offset angle. As such, any kinematic models of the robot must take into account this offset. All other joints are collinear and/or coaxial where relevant.

1.1.1 Forward Kinematics

The forward kinematics of Mo-6 can be defined using the Power of Exponentials formula discussed by Lynch and Park in Modern Robotics, [1].

As per Figure 1, define:

- h as the distance between the ground and θ_1 ;
- L_1 as the distance between θ_1 and θ_2 ;

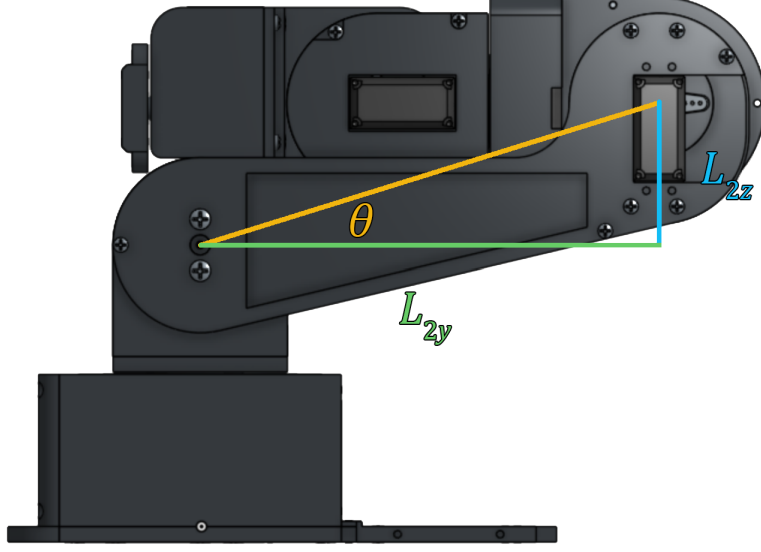


Figure 2: Diagram of y -axis and z -axis offset distances between joint 2 and joint 3.

- L_3 as the distance between θ_3 and θ_4 ;
- L_4 as the distance between θ_4 and θ_5 ;
- L_5 as the distance between θ_5 and θ_6 ;
- and L_6 as the distance between θ_6 the end-effector.

L_2 is defined as the components L_{2y} and L_{2z} , as per Figure 2. Screw axes $\mathcal{S}_n$ can then be defined for each joint as follows:

$$\mathcal{S}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ h \end{bmatrix}, \quad \mathcal{S}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ h + L_1 \end{bmatrix}, \quad \mathcal{S}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ -L_{2y} \\ h + L_1 + L_{2z} \end{bmatrix}$$

$$\mathcal{S}_4 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ -L_{2y} + L_3 \\ h + L_1 + L_{2z} \end{bmatrix}, \quad \mathcal{S}_5 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ -L_{2y} + L_3 + L_4 \\ h + L_1 + L_{2z} \end{bmatrix}, \quad \mathcal{S}_6 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ -L_{2y} + L_3 + L_4 + L_5 \\ h + L_1 + L_{2z} \end{bmatrix}$$

and the end-effector transformation at rest, M can be defined as

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -L_{2y} + L_3 + L_4 + L_5 + L_6 \\ 0 & 0 & 1 & h + L_1 + L_{2z} \\ 0 & 0 & 0 & 1 \end{bmatrix} \in SE(3)$$

which can be used in the PoE formula:

$$T(\theta) = e[\mathcal{S}_1]\theta_1 e[\mathcal{S}_2]\theta_2 e[\mathcal{S}_3]\theta_3 e[\mathcal{S}_4]\theta_4 e[\mathcal{S}_5]\theta_5 e[\mathcal{S}_6]\theta_6 M \in SE(3), \quad (1)$$

$$\theta = [\theta_1 \quad \theta_2 \quad \theta_3 \quad \theta_4 \quad \theta_5 \quad \theta_6]^T \in \mathbb{R}^6$$

Table 1 shows the physical values of the parameters used in Equation 1.

Table 1: Physical values for PoE parameters.

Parameter	Value (mm)
h	65.35
L_1	50.30
L_{2y}	179.00
L_{2z}	55.30
L_3	65.85
L_4	44.30
L_5	88.00
L_6	(Decided by chosen end-effector)

1.1.2 Workspace & Joint Limits

Table 2: Joint angle limits, listed with reference axis taken as 0° .

Joint	Reference Axis	Limit ($^\circ$)	Rest Position ($^\circ$)
θ_1	x	180° ¹	90°
θ_2	y	180° ¹	0°
θ_3	$-y$	180°	0°
θ_4	$-x$	180°	90°
θ_5	$-z$	180°	90°
θ_6	x	180°	90°

As the robot is constructed using servos for actuation, the joints are not continuous and have specific limits. These can be seen in Table 2. The reference axes and subsequent rest positions for each joint are based on the servos themselves rather than the mathematical model of Equation 1. The reference axes are defined as per the axes shown in Figure 1.

¹Technically, θ_1 and θ_2 both use the Miuzei MS24 20kg Digital Servo, which has an angle range of 270° , however for consistency, ease of use, and to simplify the construction, the servo has simply been taken as having the same 180° limit as the MG996R servos used in the other joints.

1.2 Design

References

- [1] K. M. Lynch and F. C. Park, *Modern Robotics: Mechanics, Planning, and Control*. Cambridge University Press, 2017, ISBN: 978-1-107-15630-2.